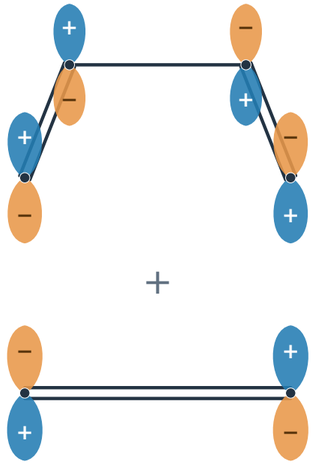
Diels–Alder [4+2] | orbital phases along the reaction

THERMALLY ALLOWED

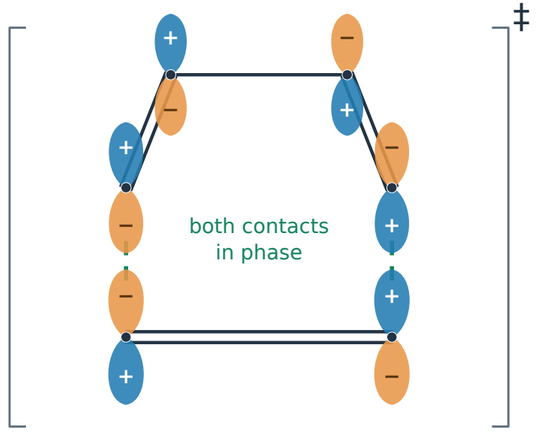
suprafacial / suprafacial

REACTANT ORBITALS	PROPOSED TRANSITION STATE	PRODUCT
Diene HOMO, ψ₂ (occupied)	Hückel circuit · 6 electrons	Product connectivity



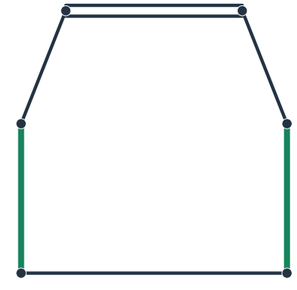
Ethene LUMO, ψ₂ (empty)

Δ



Partial bonds join the facing lobes.

Δ



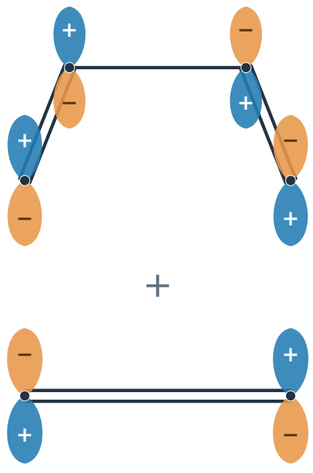
Cyclohexene

New σ bond(s) highlighted in green.

THERMALLY FORBIDDEN

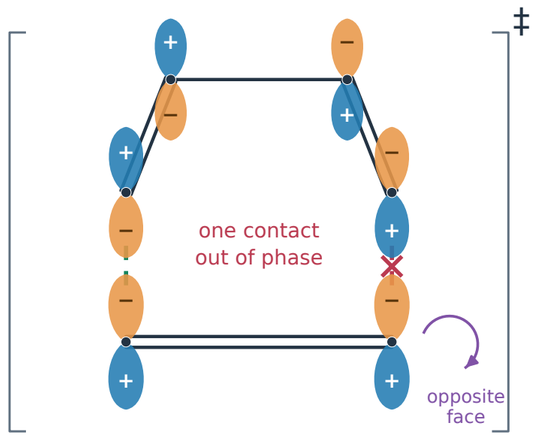
suprafacial / antarafacial

Diene HOMO, ψ₂ (occupied)	Möbius circuit · 6 electrons	Proposed product connectivity

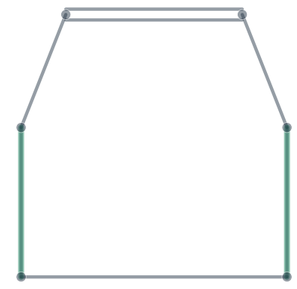


Ethene LUMO, ψ₂ (empty)

Δ



Partial bonds join the facing lobes.



Cyclohexene

Forbidden for this concerted thermal route.

+ / − orbital phase (not charge)

forming overlap / bond

phase mismatch

Cycloaddition panels show the diagnostic HOMO–LUMO pair. Changing one approach face changes the relative overlap sign.

Thermal, concerted pathways. Colored lobes are qualitative phase schematics; the middle panels are not optimized TS geometries or TS eigenorbitals.